Abhijith Neil Abraham

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Personal Profile

Ex Entrepreneur, 6+ years of ML and Software experience spanning across the entire ML, Full stack and large data lifecycle (Including Benchmarked speed of LLM inference in production, Coding Agents, large data transformations, enterprise automations) spanning major open source products, full stack applications with several users, strong research publications. Strong Leadership capabilities that led to several customer wins, along with mentoring experience for over 5000+ people.

Work Experience

Vitalops.ai *Remote*

CEO *Feb 2024 - Present*

- Led LLM inference optimization and SOTA coding agent harnesses for several LLMs for Blackbox AI (10M+ users): patched and tuned vLLM/SGLang on NVIDIA Blackwell GPUS to more than 2X benchmark speed at Artificial Analysis benchmark, sustaining 10M+ tokens/min with under replayed production load, with accuracy proven via custom eval suites. The coding agents and Inference TPS I built was benchmarked against Artifical Analysis with top 3 in the leaderboard
- Acquired paying enterprise customers across Supply Chain and Finance; shipped 5 products in 10 months owning full-stack engineering, AI Agents RAG, and model training/Inference end-to-end. Raised a pre-seed round; interviewed at Y Combinator.
- Open Source Contributor to VLLM-Metal from VLLM Project, Openvibe (1.5k stars), Datatune and Opendesk: an agentic code harness, large-scale data transformation engine, and multi-machine computer use MCP server, altogether aimed at full scale enterprise automation

Gomi.ai *Remote*

CTO *Oct 2023 - Feb 2024*

- Built and launched a full end-to-end MVP for AI-powered sales demo automation, from prototype to working product in under 4 months.

Aecmaster *Remote*

Full Stack and AI Research Engineer *March 2023 - Oct 2023*

- Led a 5-month initiative to integrate LLMs into the AEC ecosystem, delivering a full-stack version control platform with real-time 3D construction model rendering via Three.js, and a Chrome extension for handling AEC data into ChatGPT, serving customers across 3 enterprise accounts.

Autonomio *Helsinki, Finland*

Machine Learning Engineer *July 2022 - March 2023*

- Authored and open-sourced Jako, a distributed hyperparameter optimization tool that parallelizes experiments across machines, and EasyEnergy, a carbon and energy tracking tool for ML experiments, both owned end-to-end from ideation through documentation.

Saama *Chennai, India*

Machine Learning Research Engineer *Dec 2020 - June 2022*

- Published BioSimCSE at ACL's LOUHI workshop, a peer-reviewed clinical NLP venue, advancing biomedical sentence embeddings using contrastive learning.
- Pre-trained and fine-tuned biomedical models for tasks including Question Answering, and built internal tooling automating cross-matching across medical domain databases, cutting manual data reconciliation for research teams serving enterprise customers with over $400M in revenue.

Minicog *Kochi, India*

Freelance Machine Learning Engineer *May 2020 - Dec 2020*

- Built TableQA, an open-source NL-to-SQL engine with 320+ GitHub stars and 9 academic citations, and published an accompanying research paper documenting the approach.
- Co-authored a chapter published in *Advanced Deep Learning for Engineers and Scientists* (Springer, 2021), a peer-reviewed engineering book covering deep learning applications across computer vision, biometrics, and smart grids.

Education

College of Engineering, Trivandrum *Trivandrum, India*

BTech in Electrical and Electronics *August 2016 - June 2020*

- Graduated with 8.1 CGPA on a scale of 10

Skills

LLM Inference & Performance	Inference optimization, speculative decoding, quantization, GPU profiling & bottleneck analysis, Confidential Computing, load testing · vLLM, SGLang, NVIDIA Blackwell GPUs
Agent Systems	Agent harness design, multi-agent orchestration, verify-then-escalate chains, LLM-as-judge verification, tool & MCP design, context engineering, failure-mode analysis
Evaluation & Research	Eval harness design, benchmark replication, experiment design & ablation, statistical significance testing, accuracy-regression verification · Terminal-Bench, BFCL
Systems & Delivery	Distributed GPU infrastructure, full-stack engineering, large-scale data transformation, RAG, Agentic AI Engineering, secure deployment & attestation · Python, SQL, Postgres, Docker, AWS, GCP, Linux